

HAT-P-36b

R-0953

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_230220 · created 2026-07-29T14:48:05+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459995.87 +/- 0.005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.045
Transit depth (Rp/Rs)^2	2.46 +/- 1.41 [%]
Orbital Inclination (inc)	80.96 +/- 2.77
Airmass coefficient 1 (a1)	0.64 +/- 0.11
Airmass coefficient 2 (a2)	0.43 +/- 0.26
Scatter in the residuals of the lightcurve fit is	16.77 %
Best Comparison Star	#3 - [302, 370]
Optimal Method	PSF photometry
Transit Duration (day)	0.077 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.157 ± 0.045 vs 0.126 (deviation 0.69σ)
Duration fit vs published	0.077 d vs 0.0925 d (deviation 0.86σ)
Mid-time O–C	6.3 min
Depth SNR	3.5
Residual scatter	16.77 %

Light curve

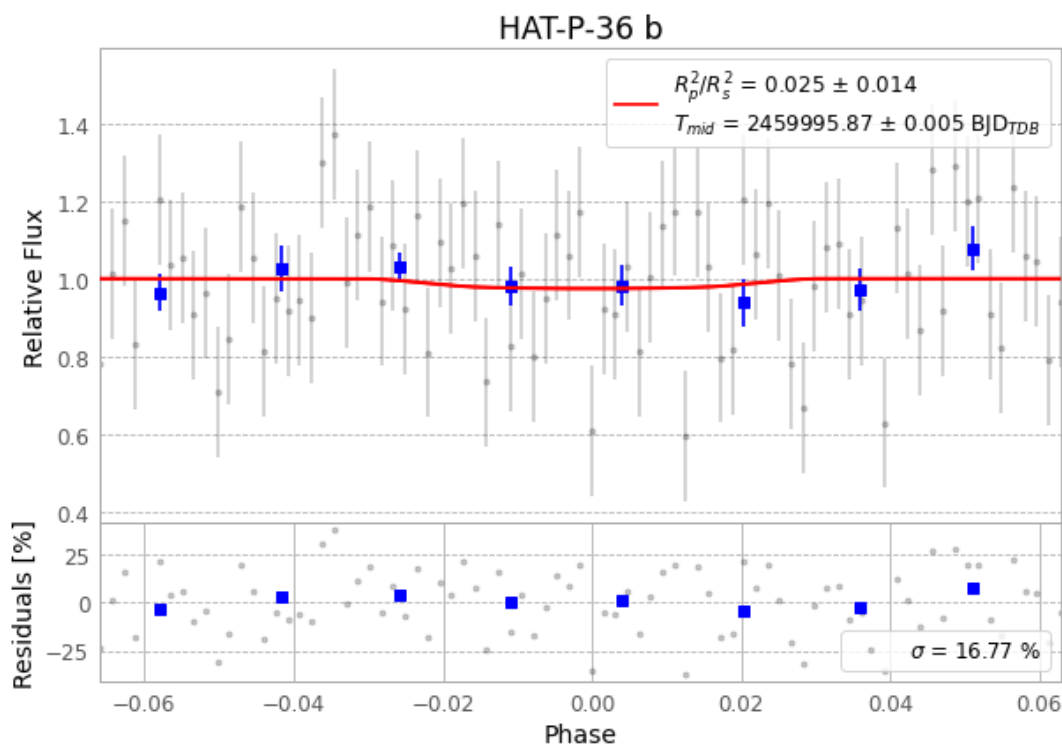


Figure 1 — FinalLightCurve_HAT-P-36 b_2023-02-19.png (SHA-256 cf7e57d66abe8cfd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [527, 264], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 d7528b3ff4ac8dfe...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

83 FITS frames (55 MB), HATP-36230220063915.FITS → HATP-36230220104515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230220.log (5b9c409007d6...), FinalLightCurve_HAT-P-36 b_2023-02-19.png (cf7e57d66abe...), FinalLightCurve_HAT-P-36 b_2023-02-19.csv (558569c6fbd6...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 14:48Z.